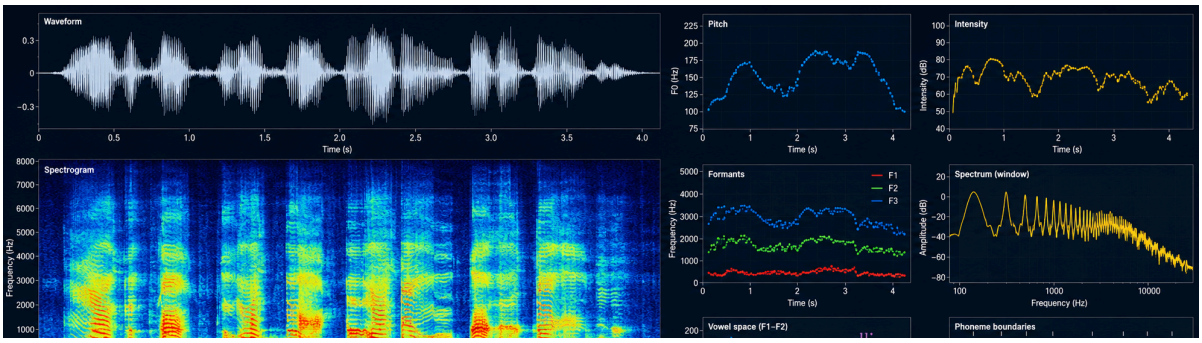
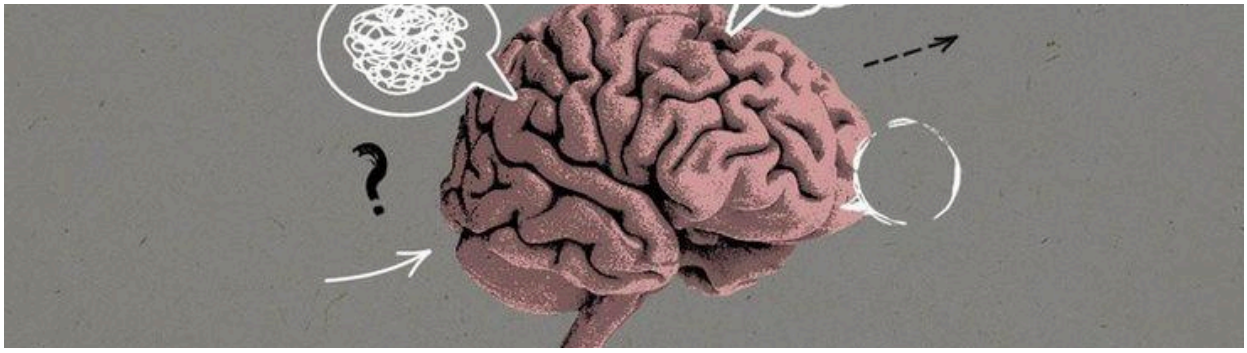


LING406: Psycholingustrics

Week	Lecture Date	Lecture Title
Week 1	8/25/2026 8/27/2026	Intro to Psycholinguistics / Things that start with "ph" Morphemes and "up"
Week 2	9/1/2026 9/3/2026	How Brains Work Language Research and Cognitive Neuroscience
Week 3	9/8/2026 9/10/2026	Linguistics and the Cognitive Revolution (Chomsky's Review of *Verbal Behavior*) Speech Errors and a Model of Speech Production I
Week 4	9/15/2026 9/17/2026	Speech Errors and a Model of Speech Production II Experimental Methods
Week 5	9/22/2026 9/24/2026	Exam 1 Review Exam 1
Week 6	9/29/2026 10/1/2026	Speech Perception I: Categorical Perception Speech Perception II: Words
Week 7	10/6/2026 10/8/2026	Experimental Studies of Sentence Processing Not listed in syllabus

Week 8	10/13/2026 10/15/2026	Aphasias Language Development
Week 9	10/20/2026 10/22/2026	More Language Development Critical Periods for L1
Week 10	10/27/2026 10/29/2026	Second Language Acquisition (LLM short paper due) Sensitive Periods for L2
Week 11	11/3/2026 11/5/2026	Exam 2 Reading
Week 12	11/10/2026 11/12/2026	Computational Models of Word Recognition More Modeling: Language Impairment and Models of the Past Tense
Week 13	11/17/2026 11/19/2026	Language and Thought Language in Other Modalities: Sign and Braille
Week 14	11/24/2026 11/26/2026	Task Analysis and Practical Issues No Class (Thanksgiving)
Week 15	12/1/2026 12/3/2026	Final Review / Catch-up In-Class Final Exam
Week 16		No class
Finals	12/15/2026	No exam (the final is given in class on 12/3)



Module II

Lecture	Slides
8	
9	
10	
11	
12	
13	
14	
Exam 2	

Lecture 11: Speech Perception I: Categorical perception

Lecture Slides & Notes
LING406 2026 - Lecture 11 Speech Perception I.pdf
Key Concepts
LING406-L11.pdf LING406_L11V.pdf LING406_Lecture11_Notes.pdf
Worked Examples / Practice Problems
Chapter Review

Lecture 12: Speech Perception II: Words

Lecture Slides & Notes
LING406 2026 - Lecture 12 Speech Perception II.pdf
Key Concepts
LING406_Lecture12_Notes.pdf LING406-L12.pdf
Worked Examples / Practice Problems
Chapter Review

Lecture 13: Aphasias

Lecture Slides & Notes
LING406 2026 - Lecture 13 Aphasias.pdf
Key Concepts
LING406_Lec13_Aphasias_Notes.pdf LING406_Lec13.pdf
Worked Examples / Practice Problems
Chapter Review

Lecture 14: Language Development 1: Speech Perception

Lecture Slides & Notes
LING406 2026 - Lecture 14 Language Development Part I.pdf
Key Concepts
LING406_Lec14_LangDevelopment_Notes.pdf LING406_Lec14.pdf
Worked Examples / Practice Problems
Chapter Review

Lecture 15: Development II: Words, Morphemes, Constructions

Lecture Slides & Notes
LING406 2026 - Lecture 15 Language Development Part II.pdf
Key Concepts
Worked Examples / Practice Problems
Chapter Review

Lecture 16: Critical Periods for a First Language (L1)

Lecture Slides & Notes
LING406 2026 - Lecture 16 Critical Periods for L1.pdf
Key Concepts
Worked Examples / Practice Problems
Chapter Review

Lecture 17: Sensitive Periods for L2 Learning

Lecture Slides & Notes
LING406 2026 - Lecture 17 Sensitive Periods for L2.pdf
Key Concepts
Worked Examples / Practice Problems
Chapter Review

Lecture 18: Second Language Acquisition

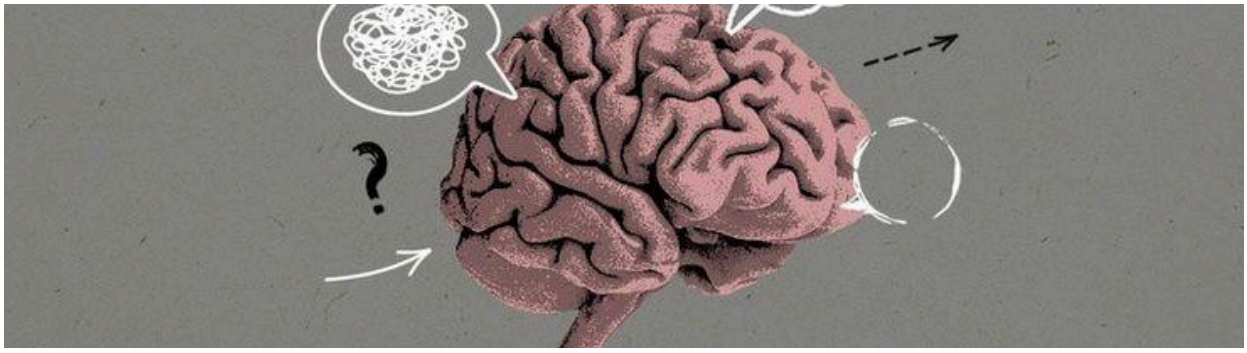
Lecture Slides & Notes
LING406 2026 - Lecture 18 Second Language Acquisition.pdf
Key Concepts
Worked Examples / Practice Problems
Chapter Review

Lecture 19: Sentence Processing

Lecture Slides & Notes
LING406 2026 - Lecture 19 Sentence Processing.pdf
Key Concepts
Worked Examples / Practice Problems
Chapter Review

EXAM II Review

Lecture Slides & Notes
LING406 2026 - Lecture 20 Midterm Review II.pdf
Key Concepts
Worked Examples / Practice Problems
Chapter Review



Module 1

Lecture	Slides
1	LING406 2026 - Lecture 1 - Things that start with ph.pdf
2	LING406 2026 - Lecture 2 - morphemes and up.pdf
3	LING406 2026 - Lecture 3 - How Brains Work.pdf
4	LING406 2026 - Lecture 4 How Brains Work.pdf
5	LING406 2026 - Lecture 5 Brains and Language.pdf
6	LING406 2026 - Lecture 6 Cognitive Revolution.pdf
7	LING406 2026 - Lecture 7 Speech Errors I.pptx
8	LING406 2026 - Lecture 8 Speech Errors II.pdf
9	LING406 2026 - Lecture 9 Experimental Design.pdf
10	LING406 2026 - Lecture 10 Midterm Review I.pdf

Exam

Lecture 1

Lecture Slides & Notes
LING406 2026 - Lecture 1-2 Things that start with ph.pdf LING406_Lecture1_Notes.pdf LING406_Lecture1_SNotes.pdf LING406_Lecture1_LNotes.pdf
Key Concepts
Worked Examples / Practice Problems
Chapter Review

Chapter notes:

- Language is orgnaized by size of the pieces it can be divided into from small to large
    - Small units are the sound that make up spoke words/letters that make up written ones
    - Larger units include words, phrases, sentences, and paragraphs or conversational turns
- Word, phrase sentences are used for both written and spoken language
    - Others are only for describing writing or only for describing speaking
- People speak in sounds NOT LETTERS
-

Arbitariness: there is no intrinsic relation between the form of an utterance (the “signifier”) and what it refers to (the “signified”)

- No intrinsic relationship between a sound and its meaning (does this include words?)
- Sounds in different languages mean different things (sometimes there is no relationship)

Productivity (or generativity)

Composability: the ability to build new words and phrases out of smaller building blocks.

- No problem with understanding all of the sentences; the sentences we hear and read are likely something that we do not encounter in day to day life.
    - E.g., happiness, sadness

Openendedness: theres no limit to the number of words or sentences we can create

Recursion: we can create complex phrases that have complex phrases inside of them

- Ex. I heard the mai heard the mary heard that maho said this movie is good
    - This movie is good - complete sentence
    - Maho said this movie is good - complete sentence

    - Sentences in sentences → embedding infinitely
    - Limit is the ability to memory (working memory)

Displacement: no spatially present; hypothetical and false hypotheticals

- Hypothesizing about things that are known to be false

Duality of patterning: languages build units by combining meaningless building blocks

- Meaningless but building things out of them from a second level

No conscious declarative knowledge; but knowledge that is able to become; unconsciously and doing it without thinking

Native speakers: who learned it at a very early age, usually form birth (and to some extend day to day life)

Experimental models:

- Segmenting a continuous stream of sounds into meaningful parts
    - Thecatis, playingintheback, yard
    - More important in the syllables
        - Segmenting the speech stream. Continuous acoustic information.
- Identification of possible and impossible words (both can be neither words)
    - Some words do not just mean ** asterisk indicate ungrammatical words
    - English contains rules, but cannot be similar to those of the other languages that take those understandings.
- Decomposition of complex words
    - No long words; open ended generativity always able to add suffix and different meanings to a word.

        - Decomposing to its components to identify its meaning
- Ability to make new forms of words that a person has never heard before
    - Language is generative; able ot understand new words even though have never experienced it.
- Distinguishment of sentences that are impossible from sentences that may be odd/weird to understand
    - Big brown dogs sleep soundly
    - ? colorless green ideas sleep furiously
- Ideas put into others peoples heads (including bizarre ones)

Linguistic competence: ability to do things that involve subconscious knowledge.

- Unable to explain the rules that underlie/underpin a person's linguistic behavior
- Linguistic competence for those who are native speakers and can switch unconsciously

Descriptive grammar to figure out the systems of subconscious mental rules that make up a native's linguistic competence

- Empirically observing, linguistics, behavior and finding patterns in what they do and what they don't say
    - Linguistic regularity

- Seeks to describe the mental rules that people have learned subconsciously without the need to have been exceptionally taught
- Only way to study language and studying

Interest in the mental representation of language and

Prescriptive grammar: is a socially embedded notion of corrent or proper use of language and it is always explicitly taught

What counts as correct or proper is almost always fetermiend by those in posiuton of autoru or power.

Correct grammar is almost always of discriminating against minoritines or people in otherwise privilege positioons.

Lecture 2

Lecture Slides & Notes
LING406 2026 - Lecture 3 Morphemes and up.pdf LING406_Lecture2_SNotes LING406_Lecture2_LNotes LING406-Lec2S.pdf LING406-Lec2L.pdf
Key Concepts
Worked Examples / Practice Problems
Chapter Review

Lecture 3

Lecture Slides & Notes
LING406 2026 - Lecture 4 How Brains Work.pdf
LING406_Lecture3_SNotes.pdf
LING406_Lecture3_LNotes.pdf
LING406_Lecture3_LNotesGPT.pdf
LING406_Lecture3_SNotesGPT.pdf
Key Concepts
Worked Examples / Practice Problems
Chapter Review

Lecture 4

Lecture Slides & Notes
LING406 2026 - Lecture 5 Brains and Language.pdf
Key Concepts
LING406_Lecture4_LNotes.pdf LING406_Lecture4_SNotes.pdf
Worked Examples / Practice Problems
Chapter Review

Lecture 6

Lecture Slides & Notes
LING406 2026 - Lecture 6 Cognitive Revolution.pdf
Key Concepts
LING406-L6_L&SNOTES.pdf
Worked Examples / Practice Problems
Chapter Review

Language as a formal and symbolic system

- Language can be described using abstract symbols and rules that specify how words combine into sentences.
- Linguistics also has strong historical connections to anthropology, including the study of language in relation to culture.
- During the 1950s and 1960s, many behaviorist psychologists opposed explanations that relied on internal mental structures or representations. The objection was to using these internal states to explain behavior.

Early computers and artificial intelligence

- Early AI programs performed tasks such as solving logic problems and playing chess.
- These programs raised the question of whether manipulating symbols according to rules could count as intelligence.

B. F. Skinner and behaviorism

- Skinner attempted to explain language through **operant conditioning**: behavior is shaped by its consequences.
- Reinforcement makes a behavior more likely to occur again.
- In *Verbal Behavior*, Skinner applied this approach to language, emphasizing the circumstances in which people speak and the consequences of their speech.
- Behaviorists emphasized observable behavior and environmental influences rather than explaining behavior through hypothetical internal mental structures.
- The note about “where a verb is” is incomplete; the broader point concerns objections to treating grammatical categories as internal things that explain behavior.

John B. Watson and the Little Albert experiment

- **John B. Watson**, rather than James Watson, conducted the experiment with **Rosalie Rayner**.
- They paired the appearance of a white rat with a frightening loud noise. Albert subsequently showed fear when presented with the rat alone.
- They also reported fear responses to some other animals and objects, interpreting this as **stimulus generalization**: a learned response extending to other stimuli.
- This illustrates **classical conditioning**, which involves learning an association between stimuli. Operant conditioning involves learning from the consequences of behavior.

Titchener and introspection

- The name is likely **Edward Titchener**.
- Titchener used **introspection**, in which trained participants reported aspects of their conscious experience.
- Behaviorists criticized reliance on these reports because private experiences were difficult to observe and verify independently.
- The specific example about reporting “how many chords” is unclear in your notes.

Internal states and experimental environments

- Behaviorists resisted using unobservable mental states as independent explanations of behavior.
- Controlled laboratory environments helped researchers study relationships between environmental conditions and observable responses.
- The comment about windowless psychology departments appears to be an anecdote about animal research facilities, or **vivariums**, rather than a general fact about psychology departments.

Skinner’s box and baby crib

- A **Skinner box** is an apparatus used to study operant conditioning in animals—for example, a rat pressing a lever to receive food.
- Skinner also designed an **air crib** to provide a comfortable, controlled environment for an infant. It was a separate invention, not a chamber for rewarding and conditioning his baby. Explanation by Skinner’s daughter

The Turing test

- A human evaluator communicates through text with a human and a computer.
- If the evaluator cannot reliably distinguish the computer from the human under the test conditions, the computer is said to pass.
- Natural language and **deception** are central: the machine must produce responses that lead the evaluator to identify it as human.
- Conversation also involves social context, making conversational ability relevant to discussions of social intelligence.
- Fluent language production raises the question of what evidence is sufficient to call a machine intelligent. Passing the test does not establish that it thinks in the same way humans do.
- Turing also proposed developing a machine with childlike learning abilities and allowing it to learn through experience.

GOFAI and the General Problem Solver

- **GOFAI** means **Good Old-Fashioned Artificial Intelligence**: an approach based on explicit symbols, rules, and procedures.
- The **1956 Dartmouth summer workshop** brought researchers together to investigate artificial intelligence. Dartmouth history
- Allen Newell and Herbert Simon were central figures in early programs for logical reasoning and problem solving.
- Their **General Problem Solver** represented a problem through a starting state, a goal state, and permitted operations.
- The broader approach was applied to formal tasks such as puzzles, mathematical proofs, and algebra problems.

Tower of Hanoi

- The goal is to move a stack of disks from one peg to another.
- Only one disk may be moved at a time.
- Only the top disk on a peg may be moved.
- A larger disk cannot be placed on a smaller disk.
- The program compares the current arrangement with the goal and selects operations or intermediate goals to reduce the difference.

- This strategy is called **means–ends analysis**. An individual move may temporarily move away from the final arrangement to make a necessary later move possible.

Emphasis on formal problems

- Researchers viewed the mind as an information-processing system, with operations that could be described in computer-like terms.
- Early computers had limited input methods, such as punch cards, and required specialized programming skills.
- Problems with clearly defined symbols, rules, and goals were easier to represent than complex sensory tasks, such as recognizing a face.
- Researchers also treated **rational, goal-directed problem solving** as a central example of intelligence. Programs could represent a goal and select procedures for reaching it.

Skinner's pigeon-guided missile project

- B. F. Skinner used **operant conditioning** to train pigeons to peck at an image of a target on a screen.
- The proposed system connected their pecks to the weapon's steering mechanism.
- If the target image shifted to one side, the pigeon would follow it and peck at its new position. These pecks would guide course corrections, bringing the target back toward the center.
- The pigeon could track changes in the target's position without understanding the weapon's purpose.
- The system was never used in combat. The project faced skepticism and was eventually discontinued as electronic guidance systems became more reliable. Describing this simply as a preference for computers over pigeons is too broad.

Chomsky @MIT machine translation program

- Cant reliably tell if a sentence is ungrammatical or grammatical → thus what is needed is a description of the grammar
    - Mastered the language to distinguis sentences from nonsentences
- If encountering a new sentence; not generalizing; on what basis is it grammatically correct

    - What is needed is a description of how sentences are generated in language
- We recognize a new item as a sentence not because it matches some familiar item in any simple way
    - Machine in the head as you process it ; produce something that you could have produced; referring the rules for producing a sentence;
- It is generated by the grammar that each individual has somehow and in some form internalized and we understand a new sentence in part because we are somehow capable of determining the proces by which this sentence is derived in this grammar
    - Children acquire essential comparable grammars of great complexity with remarkable rapidity which means that humans begnins are somehow specifically designed to do this

Behaviorism; pigenos pecking at a green light it will peck at a specific stimulus most strongly to a green light and less pecks in other colors; green light trained on is the best stimulus ; using operant conditioning

- How to respond to different situations in different behavior
- Color and number; if the pigeon will peck for 9 dots will it peck for 10 or 11; or slightly less for 20 dots
- Scale with stimulus that behaviors were used to study
- A typical example of a stimulus control for skinner would be the response to piece of music with the utterance mozart
- We cannot predict verbal behavior in terms of the stimuli in the speakers environments since we do not know wha t the current stimuli are until he responds
    - Bad theory of language according to chomsky
- You cant hope to explain why people say what they do, when they do. Thats determined by a whole atrix of social interactions, cognitive processes, previous experience → much of which has noting to do with language
- The tractable problem is to study the structure of what people say and from there to try to derive the set of rules that produce this structure

Colorless green ideas sleep furiously

- Just a bunch of words basically relying on a mechanism that tells you that a is a sentence is and b is not so cant depend on the past experience

    - The order not so much ; rather than the structure
    - Noun phrase attached to a word phrase in a particular model

Chomsky is describing lanaguate at the computational level of analysis;

- David marr is a visuall neuroscientist who talked about cognitive systems as having three levels of analysis
    - Computational
    - Algorithmic
    - Implemntational
    - Cash registers are a good ecxmple of thisl
    - Situation being solved by alorigthsm

Marr type 1 problems - computational problems

- Extracting the shape from a crowded figure
- Problems that break down clenily in these terms as type 1 problems
- Underlying strurcture common across lanaugafes and allows for ungrammatical things to occur in different languages.
    - Able to work at the computational and algorithmic levels without worrying about the implemational level (errors, memory limitations, associative learning)
- Important in the research of artificial intelligence and theoretical work in the GOFAI tradition

Type II problems

- Problems unable to be brokendown
- Devising suitable alogrithmis is not easy
- Information captured by a information system
- When understanding language its a bunch of information that is hitting us
- The whole thing in the interaction → motor control and processes of actually producing speech

Details abootu how language is implemented in the brain mgith actually be unavoaiable in the theory of language

- There is actually a whole alternative approach to pursuing language research based on just this hypothesis

Emergentism, statistical learning, parallel distributed processing

- The strong commitments on this view are at the implemntational level
- Distributed (non-hierarchical representations - like activity patterns over group neurons)
    - Instead for a node for a word or something like the physical form
- Gradual learning (like adjustment of the strength of synaptic connections)
- Focus on microstructure and more graded division between the level of analysis
- Commitments to

1990 jeff elman (1990) the core loop for modern chatbots

- Recurrent networks; how much a learning model could capture different aspects of langue performance in this framework
- All the model does is try to predict the next item it is going to see, it gradually learns to do this with experience
- Model is predicting the next item it is going to see from the current item to some memory
    - Done first for letters (m) and guesses what would come next
    - Will adjust throughout the procedure
    - Using memory from the prior state
    - Defined by the connections going in and out of them
    - Models can learn to find work boundary
        - Many occurs often
        - Unsegmented text and then predicts the next
        - Guessing the next letter; seeing many many times; likely letter would be a
        - Overtime prediction ends up learning segmentation from a word of strings
        - Finding structure but rather learning about words by predicting the next letter in a word
    - Error is low by when it gets to I
    - Where the arrow is lowest (end of words)
    - High arrow is (beginning of words)
    - Model is predicting rather the structure of words and the next thing
- Coreloop of their training is to predict the next words and fill in the banks of sentences; predicintg what is going to occur and specific locations;

Topics today:

Language as a formal and symbolic system: people were really opposed to it —> rather linguistics was connected to anthropology and expressed as something related to culture

Computer programs → artificial intelligence

- First ones solved logic / payed chess

BF skinner: no more sense to ask where a verb is

- Language understood in the stimulus response framework
- At the time the behaviorists; formal system for describing things

James watson - little albert experiment

- Fear generalizing; techniques to train animals (eg., pavlovs dog) are in humans
- Tecthner ; reporting how many people report how many chords they have → argued not real science

Models cannot make reference to this internal state

- Psychology departments dont have windows in them; designed as a vivarium space

Skinner → skinner box raising a baby in the box

- Baby rewarded for good behavior → operant conditioning

Turing test; unable to tell if its a computer then it has passed the turing test

- Early models could produce language okay but not the best
- It wasnt until a few years 5 years ish ago is that chatgpt was able to produce text quite effectively
- Models produce something that you cant tell from a model that

Depiction is important; social contect level fo social intelligence

- Chatgpt it uses an enormous amount of computation
- When are we comfortable calling it human intelligence

Building something but having it learn from the environment

GOFAI: newell and simons general problem solver - computer program

- Dartmours summer conference that brought a lot of scientists

- Specifically simon and newell was one of the first computer programs solving logic
    - Tower of hanoi ; disk goes from smallest to largest
    - Moving the disk
    - Rules of moving the disk at a time
    - No disk on top of a smaller disk
        - Also good at metiematical geometry proofs and problems
        - Getting closer to the goal by making it move?

Lecture 7

Lecture Slides & Notes
LING406 2026 - Lecture 7 Speech Errors I.ppt
Key Concepts
LING406-L7_L&SNOTES.pdf
Worked Examples / Practice Problems
Chapter Review

Lecture 8

Lecture Slides & Notes
LING406 2026 - Lecture 8 Speech Errors II.pdf
Key Concepts
LING406-L8_L&SNOTES.pdf
Worked Examples / Practice Problems
Chapter Review

Lecture 9

Lecture Slides & Notes
LING406 2026 - Lecture 9 Experimental Design.pdf
Key Concepts
LING406_L9_L&SNOTES.pdf
Worked Examples / Practice Problems
Chapter Review

Lecture 10

Lecture Slides & Notes
LING406 2026 - MIDTERM1 REVIEW.pdf
Key Concepts
LING406_MR1.pdf
Worked Examples / Practice Problems
Chapter Review